Jan 2019

Q4 Matrices, Graphs & Trees.

a) $A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 5 & 3 \\ 2 & 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 3 & -1 & 4 \\ -2 & 3 & 1 \\ 2 & 4 & -1 \end{bmatrix}$ $C = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}$

i) Identity $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

ii) $A^T = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 5 & 2 \\ 2 & 3 & 1 \end{bmatrix}$

$$A^T C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 5 & 2 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 1\times2+4\times4+2\times3 \\ 3\times2+5\times4+2\times3 \\ 2\times2+3\times4+1\times3 \end{bmatrix}$$

$3\times3 \quad \checkmark \quad 3\times1$

3×1

$$= \begin{bmatrix} 24 \\ 32 \\ 19 \end{bmatrix}$$

iii) inverse of B $\quad B^{-1} = \frac{1}{\det B} C^T$ ← Matrix of cofactors.

$\det B = b_{11}C_{11} + b_{12}C_{12} + b_{13}C_{13}$

$$B = \begin{bmatrix} 3 & -1 & 4 \\ -2 & 3 & 1 \\ 2 & 4 & -1 \end{bmatrix}$$

b_{11}: $\begin{vmatrix} 3 & 1 \\ 4 & -1 \end{vmatrix} = (3)(-1) - (1)(4) = -7$

b_{12}: $\begin{vmatrix} -2 & 1 \\ 2 & -1 \end{vmatrix} = (-2)(-1) - (1)(2) = 0$

b_{13}: $\begin{vmatrix} -2 & 3 \\ 2 & 4 \end{vmatrix} = (-2)(4) - (3)(2) = -14$

b_{21}: $\begin{vmatrix} -1 & 4 \\ 4 & -1 \end{vmatrix} = (-1)(-1) - (4)(4) = -15$

b_{22}: $\begin{vmatrix} 3 & 4 \\ 2 & -1 \end{vmatrix} = (3)(-1) - (4)(2) = -11$

b_{23}: $\begin{vmatrix} 3 & -1 \\ 2 & 4 \end{vmatrix} = (3)(4) - (-1)(2) = 14$

b_{31}: $\begin{vmatrix} -1 & 4 \\ 3 & 1 \end{vmatrix} = (-1)(1) - (4)(3) = -13$

b_{32}: $\begin{vmatrix} 3 & 4 \\ -2 & 1 \end{vmatrix} = (3)(1) - (4)(-2) = 11$

$b_{33}: \begin{vmatrix} 3 & -1 \\ -2 & 3 \end{vmatrix} = (3)(3) - (-1)(-2)$
$= 7$

cofactors.

$$M = \begin{bmatrix} -7 & 0 & -14 \\ -15 & -11 & 14 \\ -13 & 11 & 7 \end{bmatrix} \quad C = \begin{bmatrix} -7 & 0 & -14 \\ 15 & -11 & -14 \\ -13 & -11 & 7 \end{bmatrix}$$

to give

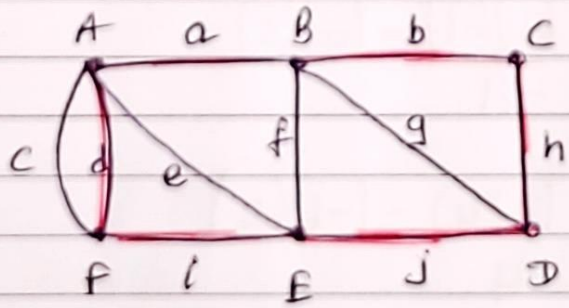
apply $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$\det B = (3)(-7) + (-1)(0) + (4)(-14)$
$= -77$

$$B^{-1} = \frac{1}{-77} \begin{bmatrix} -7 & 15 & -13 \\ 0 & -11 & -11 \\ -14 & -14 & 7 \end{bmatrix}$$

A → C^T

b)



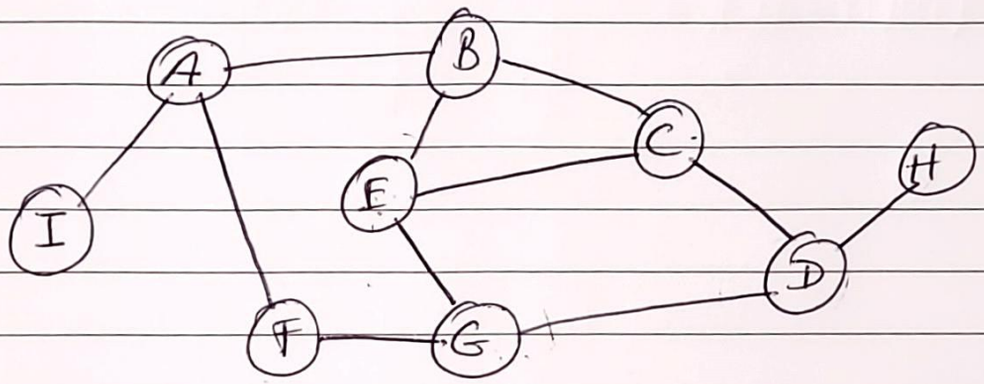
one Hamiltonian circuit = a, b, h, j, i d, (edges)

A, a, B, b, C, h, D, j E, i, f, d, A

starting & end vertex coincide ...

c) give definition/explanation of each traversal...
See Notes....

Breadth first Search.



Queue: ~~B~~ ~~F~~ ~~I~~ ~~C~~ ~~E~~ ~~G~~ ~~D~~

Output: A B F I C E G D H